

AWS Certified Developer Associate

By Stéphane Maarek



COURSE →



EXTRA PRACTICE EXAMS

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- Best of luck for the exam and happy learning!

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AWS Certified Developer Associate Course DVA-C02



Welcome! We're starting in 5 minutes

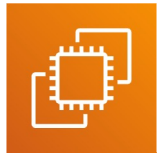
- We're going to prepare for the Certified Developer exam – DVA-C02
- It's a challenging certification, so this course will be long and interesting
- We will cover over **30** AWS services
- AWS / IT Beginners welcome! (but take your time, it's not a race)
- You don't need to be a developer to pass this exam
- Even if you've done AWS Certified Solutions Architect, don't skip lectures.



What's AWS?

- AWS (Amazon Web Services) is a Cloud Provider
- They provide you with servers and services that you can use on demand and scale easily
- AWS has revolutionized IT over time
- AWS powers some of the biggest websites in the world
 - Amazon.com
 - Netflix

What we'll learn in this course



Amazon EC2



Amazon ECR



Amazon ECS



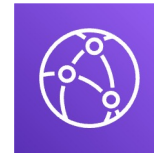
AWS Elastic
Beanstalk



AWS
Lambda



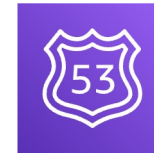
Elastic Load
Balancing



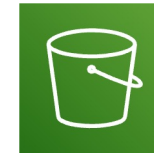
Amazon
CloudFront



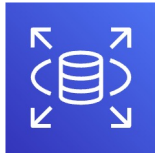
Amazon
Kinesis



Amazon
Route 53



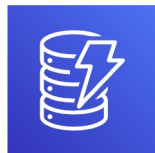
Amazon
S3



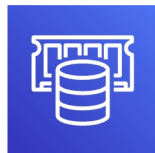
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RDS



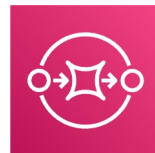
Amazon
Aurora



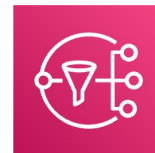
Amazon
DynamoDB



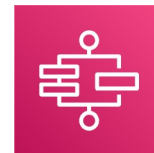
Amazon
ElastiCache



Amazon
SQS



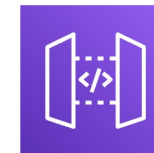
Amazon
SNS



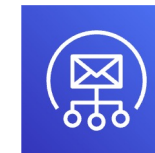
AWS Step Functions



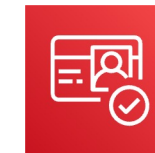
Auto Scaling



Amazon API
Gateway



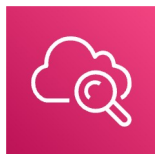
Amazon
SES



Amazon
Cognito



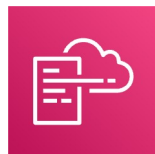
IAM



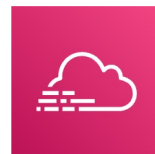
Amazon
CloudWatch



Amazon EC2
Systems Manager



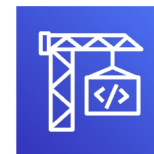
AWS
CloudFormation



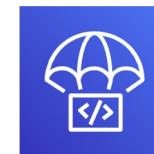
AWS
CloudTrail



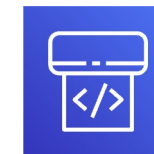
AWS
CodeCommit



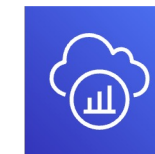
AWS
CodeBuild



AWS
CodeDeploy



AWS
CodePipeline



AWS
X-Ray



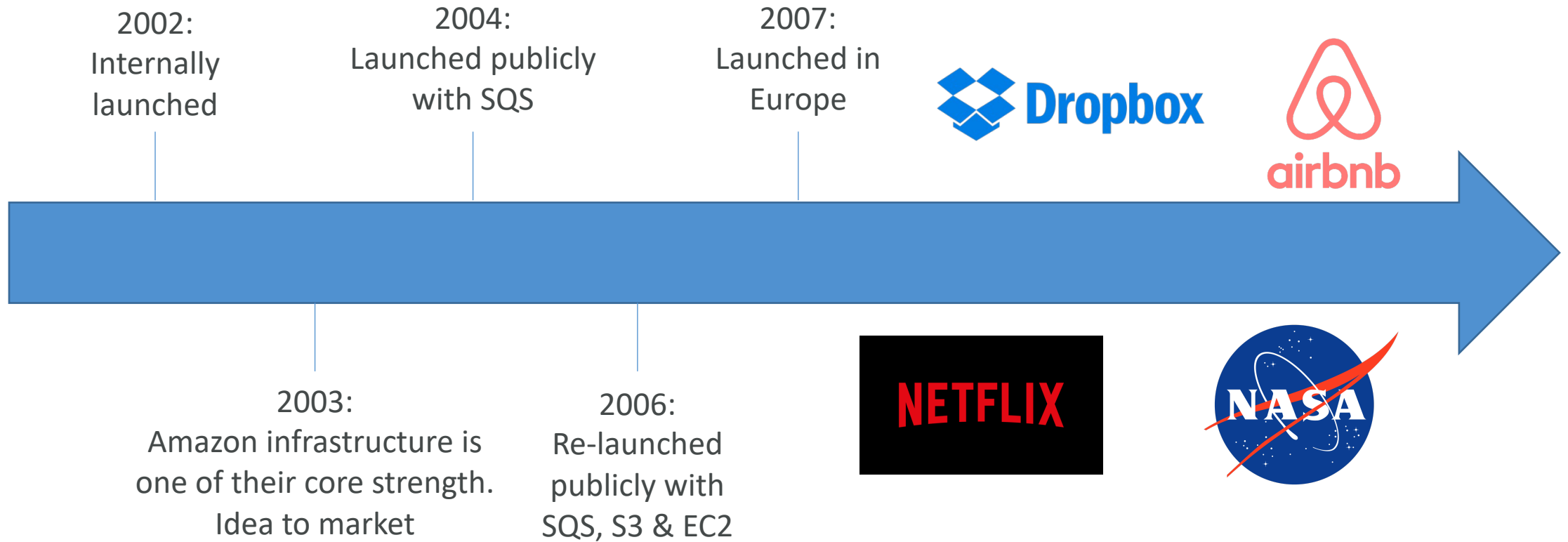
AWS KMS

Navigating the AWS spaghetti bowl



Getting started with AWS

AWS Cloud History



AWS Cloud Number Facts

- In 2023, AWS had \$90 billion in annual revenue
- AWS accounts for 31% of the market in Q1 2024 (Microsoft is 2nd with 25%)
- Pioneer and Leader of the AWS Cloud Market for the 13th consecutive year
- Over 1,000,000 active users

Figure 1: Magic Quadrant for Strategic Cloud Platform Services



Gartner Magic Quadrant

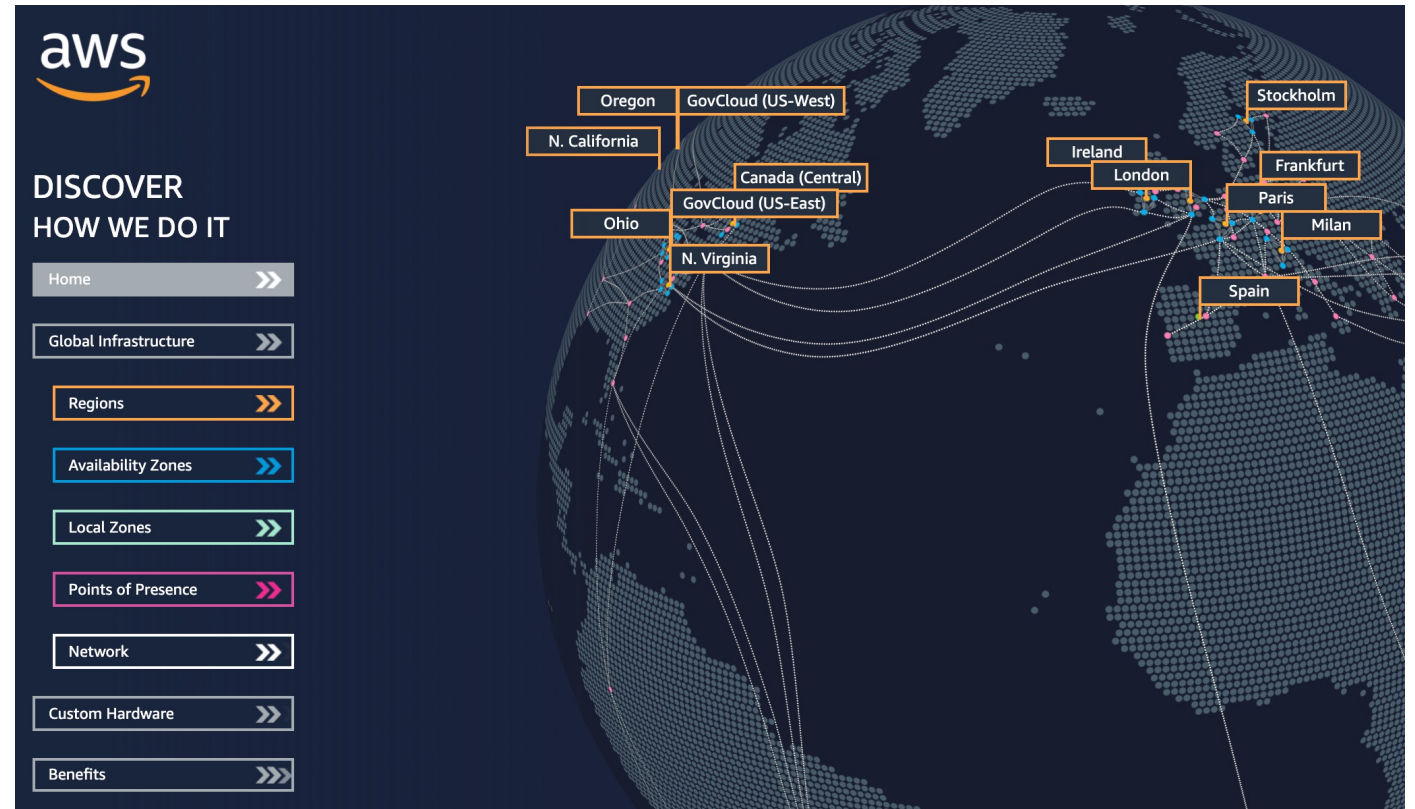
AWS Cloud Use Cases

- AWS enables you to build sophisticated, scalable applications
- Applicable to a diverse set of industries
- Use cases include
 - Enterprise IT, Backup & Storage, Big Data analytics
 - Website hosting, Mobile & Social Apps
 - Gaming



AWS Global Infrastructure

- AWS Regions
- AWS Availability Zones
- AWS Data Centers
- AWS Edge Locations / Points of Presence
- <https://infrastructure.aws/>



AWS Regions

- AWS has **Regions** all around the world
- Names can be us-east-1, eu-west-3...
- A region is a **cluster** of data centers
- Most AWS services are region-scoped



<https://aws.amazon.com/about-aws/global-infrastructure/>

US East (N. Virginia) us-east-1

US East (Ohio) us-east-2

US West (N. California) us-west-1

US West (Oregon) us-west-2

Africa (Cape Town) af-south-1

Asia Pacific (Hong Kong) ap-east-1

Asia Pacific (Mumbai) ap-south-1

Asia Pacific (Seoul) ap-northeast-2

Asia Pacific (Singapore) ap-southeast-1

Asia Pacific (Sydney) ap-southeast-2

Asia Pacific (Tokyo) ap-northeast-1

Canada (Central) ca-central-1

Europe (Frankfurt) eu-central-1

Europe (Ireland) eu-west-1

Europe (London) eu-west-2

Europe (Paris) eu-west-3

Europe (Stockholm) eu-north-1

Middle East (Bahrain) me-south-1

South America (São Paulo) sa-east-1

How to choose an AWS Region?

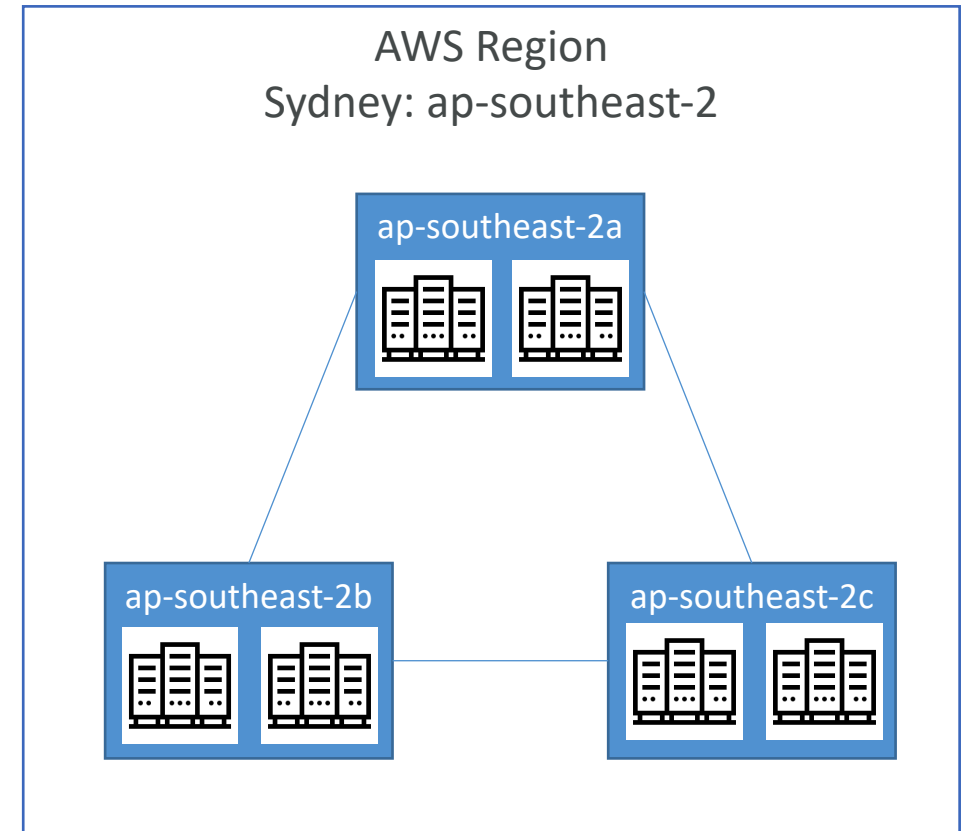
If you need to launch a new application, where should you do it?



- **Compliance** with data governance and legal requirements: data never leaves a region without your explicit permission
- **Proximity** to customers: reduced latency
- **Available services** within a Region: new services and new features aren't available in every Region
- **Pricing**: pricing varies region to region and is transparent in the service pricing page

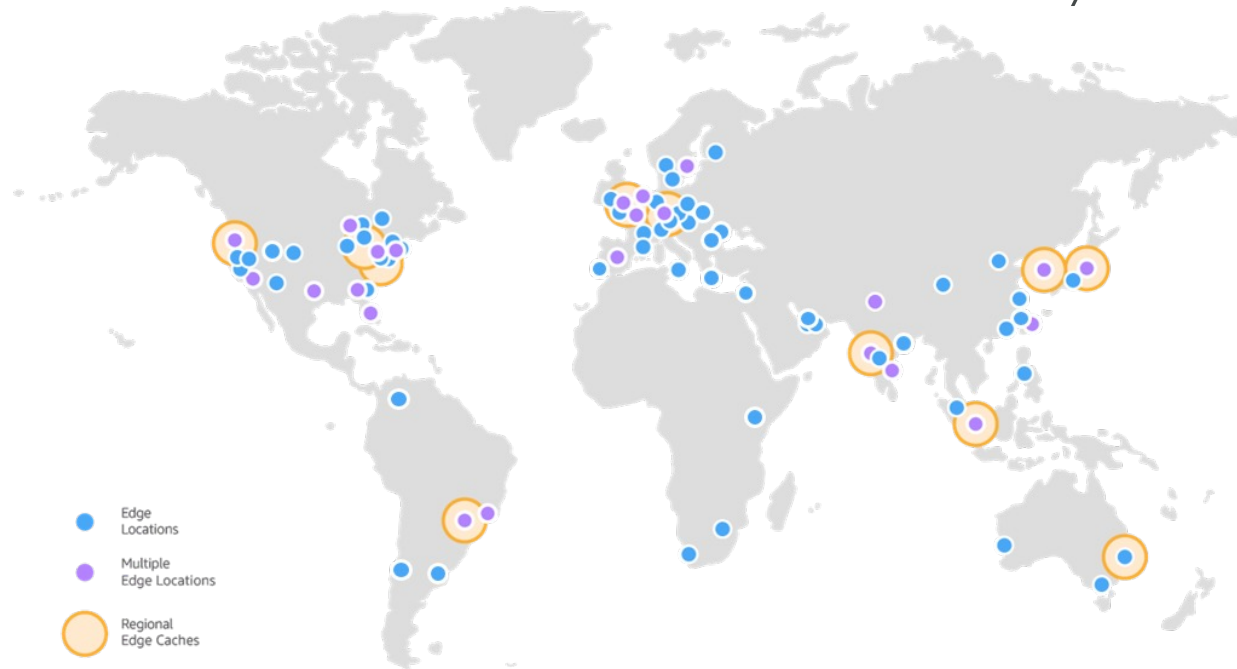
AWS Availability Zones

- Each region has many availability zones (usually 3, min is 3, max is 6). Example:
 - ap-southeast-2a
 - ap-southeast-2b
 - ap-southeast-2c
- Each availability zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity
- They're separate from each other, so that they're isolated from disasters
- They're connected with high bandwidth, ultra-low latency networking



AWS Points of Presence (Edge Locations)

- Amazon has 400+ Points of Presence (400+ Edge Locations & 10+ Regional Caches) in 90+ cities across 40+ countries
- Content is delivered to end users with lower latency



<https://aws.amazon.com/cloudfront/features/>



Tour of the AWS Console

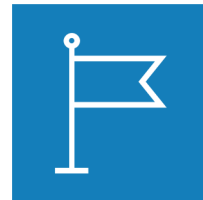
- **AWS has Global Services:**

- Identity and Access Management (IAM)
- Route 53 (DNS service)
- CloudFront (Content Delivery Network)
- WAF (Web Application Firewall)



- **Most AWS services are Region-scoped:**

- Amazon EC2 (Infrastructure as a Service)
- Elastic Beanstalk (Platform as a Service)
- Lambda (Function as a Service)
- Rekognition (Software as a Service)



- **Region Table:** <https://aws.amazon.com/about-aws/global-infrastructure/regional-product-services>

AWS Identity & Access Management (AWS IAM)